TITANIC DATASET

Exploratory Data Analysis Report

Project 1 | SyntecxHub Data Science Series | 2024

1. Executive Summary

Dataset Overview

The RMS Titanic sank on 15 April 1912, killing 1,502 of 2,224 passengers and crew. This analysis explores the training set of 891 passengers from the Kaggle Titanic competition, uncovering which demographic factors most strongly predicted survival.

891 passengers • 342 survived (38.4%) • 549 perished (61.6%) • 12 features

2. Key Metrics at a Glance

Metric	Value	Insight
Overall Survival Rate	**38.4%**	342 of 891 passengers survived
Female Survival Rate	**74.2%**	"Women & children first" policy confirmed
Male Survival Rate	**18.9%**	Only 1 in 5 male passengers survived
1st Class Survival Rate	**63.0%**	Best cabin access to lifeboats
2nd Class Survival Rate	**47.3%**	Moderate survival advantage
3rd Class Survival Rate	**24.2%**	Lower decks made evacuation harder
Child Survival Rate	**58.0%**	Highest among all age groups
Senior Survival Rate	**22.7%**	Lowest among age groups
1st Class Female	**96.8%**	Near-complete survival
3rd Class Male	**13.5%**	Most vulnerable group

3. Detailed Insights

Insight 1 — Gender Was the Strongest Predictor

Sex was the most powerful predictor of survival in the Titanic dataset, with a correlation coefficient of +0.54 with survival. The "women and children first" maritime evacuation protocol dramatically influenced outcomes.

- Females survived at 74.2% vs males at only 18.9% — nearly a 4x difference.
- 314 female passengers were aboard; 233 survived (74.2%).
- 577 male passengers were aboard; only 109 survived (18.9%).
- Even in 3rd class, female survival (50%) far exceeded male (13.5%).

Insight 2 — Passenger Class Determined Life Chances

Socioeconomic class, measured through ticket class, was the second most impactful variable. First-class cabins were located on upper decks with easier lifeboat access, while third-class passengers faced structural barriers to evacuation.

- 1st class: 63.0% survival | 2nd class: 47.3% | 3rd class: 24.2%
- 1st class females: 96.8% survival — the highest of any group.
- 3rd class males: 13.5% survival — the lowest of any gender-class group.
- Fare correlated positively with survival (r = +0.26), reflecting class and deck position.
- Cherbourg passengers (often wealthier 1st class) had 55.4% survival vs Southampton at 33.7%.

Insight 3 — Children Had Priority; Elderly Faced the Hardest Odds

Age buckets revealed a clear pattern aligned with the evacuation priority protocol. Children received explicit priority in boarding lifeboats.

- Children (0–12): 58.0% survival — highest among all age groups.
- Teens (13–18): 42.9% survival — benefited from youth priority.
- Adults (19–35): 38.3% — close to the overall average.
- Middle-aged (36–60): 40.0% — slightly above average.
- Seniors (60+): 22.7% — significantly below average; mobility limitations likely a factor.
- The median age of survivors (28) was slightly lower than non-survivors (30).

Insight 4 — Family Size Had a Survival "Sweet Spot"

- Solo travelers (FamilySize = 1): 30.4% survival — traveling alone was a disadvantage.
- Small families (2–4 members): 57% survival — best outcomes; could assist each other.
- Large families (5–6 members): ~33% — coordination became harder.
- Very large families (7+ members): below 20% — chaos in large group evacuation.

4. Data Quality Notes

Feature	Missing %	Impact	Treatment
Cabin	77%	High — deck location matters	Extract deck letter; use Has_Cabin flag
Age	20%	Moderate — age is key predictor	Imputed via Pclass + Sex group medians
Embarked	0.2%	Low — only 2 missing rows	Filled with mode (Southampton)

5. Conclusions & Recommendations

The Titanic disaster was not random — it was deeply structured by social hierarchies. Three factors overwhelmingly determined survival:

- SEX: Gender was the single strongest factor. The evacuation protocol prioritized women.
- CLASS: Socioeconomic class shaped both physical access (deck position) and social priority during evacuation.
- AGE: Children received explicit protection; the elderly faced the most adverse conditions.

For predictive modelling, the recommended feature set is: Sex, Pclass, Age, Fare, FamilySize, Embarked, and engineered features (AgeBucket, IsAlone, Has_Cabin). A Random Forest or Gradient Boosting classifier trained on these features is expected to achieve 80–83% accuracy on the test set.

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